Instructions

This mock mirrors the BI3 flow: aptitude + reasoning, then DE technical MCQs, then SQL and coding.

This is Mock 3 of 3; each uses a different question set.

Section 1 - Aptitude

Q1. If 3 pens cost Rs. 45, then 7 pens cost:

(A) Rs. 90
(B) Rs. 105 ✔
(C) Rs. 100
(D) Rs. 120

Answer: (B) Rs. 105

Authenticity: Community Reported

Q2. A number increased by 10% becomes 220. The original number:

(A) 200 ✔
(B) 210
(C) 180
(D) 190

Answer: (A) 200

Authenticity: Community Reported

Q3. 40% of 150:

(A) 50
(B) 60 ✔
(C) 70
(D) 45

Answer: (B) 60

Authenticity: Research-Backed Company Style

Q4. 4 workers build a wall in 6 days. 3 workers take:

(A) 8 days ✔
(B) 7 days
(C) 9 days
(D) 6 days

Answer: (A) 8 days

Authenticity: Community Reported

Q5. 8 kg of rice costs Rs. 320. Cost of 5 kg:

(A) Rs. 180
(B) Rs. 200 ✔
(C) Rs. 220
(D) Rs. 160

Answer: (B) Rs. 200

Authenticity: Community Reported

Section 2 - Reasoning

Q1. Next term: 5, 10, 17, 26, ?

(A) 35
(B) 37 ✔
(C) 36
(D) 40

Answer: (B) 37

Authenticity: Frequently Repeated

Q2. Next term: 1, 1, 2, 3, 5, ?

(A) 6
(B) 7
(C) 8 ✔
(D) 9

Answer: (C) 8

Authenticity: Frequently Repeated

Q3. 'He is the son of my mother's only son,' says Meena. He is her:

(A) Brother
(B) Nephew ✔
(C) Son
(D) Cousin

Answer: (B) Nephew

Authenticity: Community Reported

Q4. Facing east, turn 90 deg clockwise. Now facing:

(A) North
(B) South ✔
(C) West
(D) East

Answer: (B) South

Authenticity: Community Reported

Section 3 - Data Engineering Technical

Q1. A PRIMARY KEY guarantees values are:

(A) Sorted
(B) Unique and not null ✔
(C) Indexed only
(D) Foreign

Answer: (B) Unique and not null

Authenticity: Community Reported

Q2. Which is an aggregate function?

(A) UPPER
(B) LENGTH
(C) AVG ✔
(D) SUBSTR

Answer: (C) AVG

Authenticity: Community Reported

Q3. A snowflake schema has dimension tables that are:

(A) Denormalized
(B) Normalized into sub-dimensions ✔
(C) Removed
(D) Combined

Answer: (B) Normalized into sub-dimensions

Authenticity: Community Reported

Q4. A data warehouse stores data that is:

(A) Real-time only
(B) Subject-oriented, integrated, historical ✔
(C) Unstructured only
(D) Temporary

Answer: (B) Subject-oriented, integrated, historical

Authenticity: Community Reported

Q5. Which model handles data in real time as it arrives?

(A) Batch
(B) Stream processing ✔
(C) Manual
(D) Nightly

Answer: (B) Stream processing

Authenticity: Community Reported

Q6. Partitioning a large table helps by:

(A) Increasing cost
(B) Improving query performance via pruning ✔
(C) Removing indexes
(D) Preventing joins

Answer: (B) Improving query performance via pruning

Authenticity: Community Reported

Q7. A data lake stores:

(A) Only relational data
(B) Raw data in many formats ✔
(C) Only reports
(D) Only images

Answer: (B) Raw data in many formats

Authenticity: Community Reported

Q8. pandas operation like a SQL join:

(A) concat()

(B) merge() ✔

(C) append()

(D) pivot()

Answer: (B) merge()

Authenticity: Community Reported

Section 4 - SQL

Problem: Total Sales per Region

Total sales per region, ordered by total descending.

```
SELECT region, SUM(amount) AS total_sales
FROM sales
GROUP BY region
ORDER BY total_sales DESC;
```

Expected Output

```
region|total_sales
South|300
North|250
East|50
```

Section 5 - Coding / Python

Problem 1: Valid Anagram

Difficulty: ***Easy*** | *Pattern:* ***Frequency Counting*** | *Authenticity:* ***Community Reported***

Problem Statement

Decide whether two strings are anagrams.

Input Format

Line 1: first string. Line 2: second string.

Output Format

True or False.

Constraints

1 <= length <= 10^4

Sample Input

```
listen
silent
```

Sample Output

```
True
```

Python Solution

```
def is_anagram(a, b):
    if len(a) != len(b):
        return False
    counts = {}
    for ch in a:
        counts[ch] = counts.get(ch, 0) + 1
    for ch in b:
        if counts.get(ch, 0) == 0:
            return False
        counts[ch] -= 1
    return True

print(is_anagram('listen', 'silent'))
```

Step-by-Step Explanation

Count first string, decrement for second; all cancel -> anagram.

Dry Run

```
listen/silent letters cancel -> True
```

Time Complexity: **O(N)** | Space Complexity: **O(1)**

Problem 2: Check Prime

*Difficulty: **Easy*** | *Pattern: **Trial Division to sqrt(n)*** | *Authenticity: **Community Reported***

Problem Statement

Decide whether n is prime.

Input Format

A single integer n.

Output Format

True or False.

Constraints

1 <= n <= 10^9

Sample Input

```
29
```

Sample Output

```
True
```

Python Solution

```
def is_prime(n):
    if n < 2:
        return False
    i = 2
    while i * i <= n:
        if n % i == 0:
            return False
        i += 1
    return True
```

```
print(is_prime(29))
```

Step-by-Step Explanation

No divisor up to sqrt(n) means prime; a larger factor pairs a smaller one.

Dry Run

```
29: no divisor up to 5 -> True
```

Time Complexity: **O(sqrt N)** | Space Complexity: **O(1)**

BI3 Technologies

Full-Length Mock Test 3

Applied Role(s):

Data Engineer - Trainee

Online Assessment Preparation Material

Version 1.1 • Last Updated: July 2026

Every question carries an authenticity label. See the legend on the next page.

Table of Contents

Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.